The DICTO Truncated-Icosahedral Hecatodihedron

A convex solid of 102 faces: the truncated icosahedron with a low pyramid on each hexagon, flattened so that neighbouring triangles become rhombi



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Version 2 · 16 October 2026 · DICTO DOI: 10.5281/zenodo.23360857 (all versions) · Licence CC BY 4.0 · Data, renders and checks: papers/tih at https://github.com/DICTOR-Master/kaleidohedra.
The solid is a stage of our Spheres of Dharma, found by building it, and the name is ours. Computation and checking were done with Claude (Anthropic); the exact relation to the joined truncated icosahedron (§3) was found in a prior-art search and placed in this version on the advice of Claude Fable. **Version 2** corrects “60 pairs” to 30, adds §3 (what the solid is, exactly) and §4 (the DICTO Hecatogdoecontahedron), and revises the status (§5).

Abstract. We describe a convex polyhedron with 102 faces, 80 corners and 180 edges that arises as the seventh stage of our Spheres of Dharma, a cycle of nested icosahedral shells. It is the truncated icosahedron with a low hexagonal pyramid (a *Shield*) on each of its 20 hexagons, at the one height at which the two triangles of neighbouring Shields across every hexagon–hexagon edge lie in a single plane. Its faces are 12 regular pentagons, 30 rhombi and 60 isosceles triangles; it has only two edge angles. Combinatorially it is Conway's hexakis truncated icosahedron k6tI (132 faces) with 30 pairs of coplanar triangles merged into 30 rhombi. Exactly, its 20 Shield apices are vertices of the canonical pentakis dodecahedron, so it is the joined truncated icosahedron jtI with its 12 five-fold corners cut off at the pentagon planes: an unnamed solid derived from a known one. The rhombi and triangles all touch one sphere, the truncated icosahedron's midsphere; the pentagons lie a little inside it, so the solid has no insphere. With its polar, the DICTO Ogdoecontahedron, it forms a self-dual compound of 182 faces, the DICTO Hecatogdoecontahedron, whose core and hull are those of the classical truncated icosahedron–pentakis dodecahedron compound. We name it the **DICTO Truncated-Icosahedral Hecatodihedron** (*hecatodi-*, one hundred and two). This corrects an earlier statement of ours that all 102 faces touch one sphere.

1. Origin

The Spheres of Dharma (DISCOVERIES #23) grow an icosahedron, layer by layer, through the icosidodecahedron, the dodecahedron, the truncated icosahedron and the rhombic triacontahedron, back to an icosahedron φ^3 larger. Taking each closed stage as one solid (our paper *Capturing the In-betweens*, DICTO DOI: 10.5281/zenodo.23345502), stage 7, the truncated icosahedron with a Shield on every hexagon, turned out to be convex, with fewer faces than its pieces suggest: 102. It is a station of our DICTO-TIC world (Transitional Icosahedral Cycles) and a shape in Polyhedraverse's DICTO Transitional Shells.

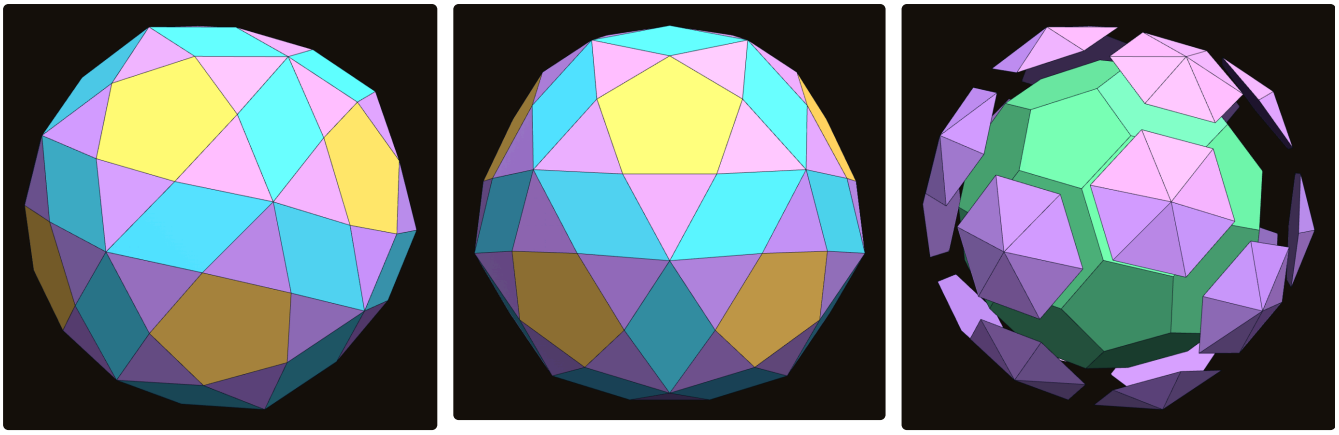


Fig. 1–3. (1) The Hecatodihedron: 12 regular pentagons (gold), 30 rhombi (cyan), 60 triangles (wisteria). (2) Down a five-fold axis. (3) The truncated icosahedron (green) and its 20 Shields, pulled apart.

2. The solid

Start from the truncated icosahedron of edge $1/\varphi$ (the scale of the Dharma build). On each hexagon stand a hexagonal pyramid whose apex lies on the hexagon's three-fold axis. Raise the apices together: there is one height at which the triangle of one Shield and the triangle of its neighbour, across each edge where two hexagons meet, lie in one plane. At that height these 60 triangles pair up into 30 rhombi, and the 60 triangles across the hexagon–pentagon edges stay triangles. The 20 apices are then the corners of a regular dodecahedron.

Faces	Number	Shape	Distance from the centre
Regular pentagons	12	side $1/\varphi = 0.61803$	1.4384
Rhombi	30	sides 0.65097, angles 56.68° and 123.32°	1.5
Isosceles triangles	60	sides 0.61803, 0.65097, 0.65097	1.5

Corners: 80, the 60 of the truncated icosahedron (all at distance 1.5315) and the 20 Shield apices (at 1.6057), which form a regular dodecahedron of edge $3/\varphi^2 = 1.14590$. Edges: 180, with only two edge angles, 159.45° and 163.53°. Euler's formula: $80 - 180 + 102 = 2$. Volume 14.4042 at this scale.

A sphere, but not an insphere. The 30 rhombi and 60 triangles all lie at distance 1.5 from the centre, so 90 of the 102 faces touch one sphere. The 12 pentagons, the truncated icosahedron's own, lie at 1.4384, a little closer; the solid therefore has no insphere. Our earlier paper said that all 102 faces touch one sphere; that was wrong, and is corrected in its version 3.

3. What the solid is, exactly

The truncated icosahedron's dual, the pentakis dodecahedron, taken canonically (as the truncated icosahedron's polar in its own midsphere, radius 3/2 at this scale), has 12 five-fold and 20 three-fold vertices. Exactly, in $\mathbb{Q}(\sqrt{5})$ (`data/tih_v2_check.py`):

- **The 20 Shield apices are the pentakis dodecahedron's 20 three-fold vertices.** So the Hecatodihedron is the convex hull of the truncated icosahedron and these 20 points.
- **Its face planes are exactly the 90 face planes of the joined truncated icosahedron jtI** (the convex hull of the truncated icosahedron and the whole pentakis dodecahedron, topologically the rhombic enneacontahedron) **together with the**

truncated icosahedron's 12 pentagon planes: 102 = 90 + 12. So the Hecatodihedron is the jtI with its 12 five-fold corners cut off at the pentagon planes.

This explains the solid. The Shield height at which neighbouring triangles flatten is the joined height of the jtI construction, in which pyramids on the truncated icosahedron's faces are raised until their faces become coplanar; here only the hexagons carry pyramids. The 90 faces at distance 1.5 are faces of the jtI, which touch the midsphere at the truncated icosahedron's edge midpoints; the 12 pentagons are the truncated icosahedron's own, at 1.4384.

4. The DICTO Hecatogdoecontahedron

The Hecatodihedron's polar in the same sphere is the DICTO Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688), and the two interpenetrate like the two tetrahedra of the stella octangula. DICTO named the compound the **DICTO Hecatogdoecontahedron** (102 + 80, after the icosidodecahedron's 20 + 12): 182 faces and 182 corners, self-dual, its polar swapping the two solids. Exactly: the Ogdoecontahedron's 90 sphere corners are the truncated icosahedron's 90 edge midpoints, and kiss the Hecatodihedron's 90 tangent faces; its 12 five-fold corners poke out through the pentagons; the Hecatodihedron's 80 corners poke out of the Ogdoecontahedron. The compound's hull has exactly the corners of the truncated icosahedron and the pentakis dodecahedron, so it is the jtI, and its common core is the rectified truncated icosahedron. These are the hull and core of the classical truncated icosahedron–pentakis dodecahedron compound: the new pair re-partitions a classical compound, as the icosahedron–dodecahedron compound has the icosidodecahedron and the rhombic triacontahedron as core and hull. By the same polarity the Ogdoecontahedron is exactly the pentakis dodecahedron with its 20 three-fold corners cut off at the truncated icosahedron's hexagon planes.

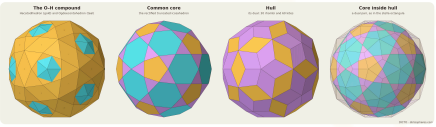


Fig. 4. The DICTO Hecatogdoecontahedron: the Hecatodihedron (gold) and the Ogdoecontahedron (teal) interlocked; their common core, the rectified truncated icosahedron; the hull, the jtI; the core inside the hull. Drawn from exact solids.

5. Relation to known solids, and status

Combinatorially, a pyramid on each hexagon of the truncated icosahedron is Conway's hexakis truncated icosahedron, k6tI, with 80 corners and 132 triangular-and-pentagonal faces. The Hecatodihedron is the particular, flattened realisation in which 30 pairs of its triangles are coplanar, so that it has 102 faces and 180 edges. It is not an Archimedean, Catalan or Johnson solid. As §3 shows, it is the joined truncated icosahedron cut at the pentagon planes: an unnamed solid derived from a known one. We did not find it named in the catalogues listed below.

Where we searched (2026-10-13 and 2026-10-14), and the nearest solids we found:

- **Conway notation** (Conway; Hart's *Conway Notation for Polyhedra* [3, 4]): the hexakis truncated icosahedron k6tI has 80 corners and 132 faces; the pentakis truncated icosahedron k5tI 72 corners and 132 faces; the hexapentakis truncated icosahedron ktI 92 corners and 180 faces [5]. None has 102 faces; ours is k6tI flattened.

- **Johnson solids and their near misses** [6, 7, 8]: not a Johnson solid (its rhombi and triangles are not regular); the nearest near miss in face content is Kaplan's rectified truncated icosahedron (92 faces: 60 isosceles triangles, 12 regular pentagons, 20 hexagons) [8], which has hexagons where ours has rhombi.
- **Archimedean and Catalan solids** and their compounds and duals in Wolfram MathWorld and the PolyhedronData collection [9]: the truncated icosahedron, the pentakis dodecahedron and their compound; nothing with 102 faces.
- **The Polytope Wiki** (Miraheze) [10] and named polyhedra with 80 or 92 corners found by web searches for these counts and face lists: no match.
- **Added in Version 2** (2026-10-16): the rhombic enneacontahedron and its Conway form jtI [11], the truncated icosahedron–pentakis dodecahedron compound [12], and D. McCooey's Visual Polyhedra (Archimedean–Catalan hulls, including the joined truncated icosahedron; rectified Archimedean solids) [13]. These give the jtI and the rectified truncated icosahedron, from which §3–4 follow; the Hecatodihedron itself (102 faces) was not found among them.

That is a report of where we looked, not a claim that the solid is unknown.

The name: *hecatodi-* for its 102 faces (Greek *hekaton*, 100, and *di*, 2), as in the hecatonicosachoron, the 120-cell; *truncated-icosahedral* for the solid it grows from.

6. Reproducibility

Included: `data/tih_v2_check.py` and `data/tih-v2-check.txt` (§3–4, exact in $\mathbb{Q}(\sqrt{5})$; also in studies/2026-10-16-progenitors), `data/tih_check.py` and `data/tih-checks.json` (face kinds, distances, corners, edges, edge angles, volume and the dodecahedron of apices, computed from Polyhedraverse's shape TRANS_DHARMA_80), and `data/tih_fig.py` (the figures). The solid can be opened in Polyhedraverse (DICTO Transitional Shells) and seen arriving in DICTO-TIC, Loop 2.

References

1. DICTO, Capturing the In-betweens: a study of transitional polyhedral shells (2026), DICTO DOI: 10.5281/zenodo.23345502 (version 3 corrects the insphere statement).
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5. Hexapentakis truncated icosahedron (with the k6tI and k5tI forms listed alongside), Wikipedia, https://en.wikipedia.org/wiki/Hexapentakis_truncated_icosahedron.
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